# Training Day 10

## 1. Topics Covered and Brief Explanation

1. Introduction to Exploratory Data Analysis (EDA)

Exploratory Data Analysis is the process of examining and understanding a dataset before applying machine learning algorithms. It helps identify patterns, trends, relationships and unusual values in the data.

2. Importance of EDA

Studied how EDA helps in:

- Understanding the structure of datasets.
- Identifying missing and duplicate values.
- Finding patterns and relationships between variables.
- Detecting unusual or extreme values.
- Making better decisions during model development.

3. Basic Statistical Analysis

Learned about common statistical measures used to understand numerical data:

- Mean – Average value of a dataset.
- Median – Middle value of sorted data.
- Mode – Most frequently occurring value.
- Minimum and Maximum – Smallest and largest values.
- Standard Deviation – Measures the spread of data values.

4. Introduction to Data Visualization

Data visualization is the process of representing information through graphs and charts. It makes complex datasets easier to understand and interpret.

Studied different types of charts:

- Bar Chart: Used to compare different categories.
- Line Chart: Used to display trends over time.
- Histogram: Used to understand the distribution of numerical data.
- Scatter Plot: Used to identify relationships between two numerical variables.

5. Introduction to Matplotlib

Matplotlib is a Python library used to create graphs and visual representations of data. It helps in analysing datasets through different types of plots.

## 2. Tools and Technologies Used

| Tool | Purpose |
|---|---|
| Python | Performing data analysis. |
| Pandas | Exploring and summarizing datasets. |
| NumPy | Performing numerical calculations. |
| Matplotlib | Creating graphs and charts. |
| Google Colab | Writing and executing Python programs. |

## 3. Practical Learning / Code Examples

Example 1: Basic Statistical Analysis

```
import pandas as pd
data = {
   "Marks": [65, 75, 85, 90, 95]}
df = pd.DataFrame(data)
print("Mean:", df["Marks"].mean())
print("Median:", df["Marks"].median())
print("Minimum:", df["Marks"].min())
print("Maximum:", df["Marks"].max())
```

Example 2: Creating a Bar Chart

```
import matplotlib.pyplot as plt
names = ["Aman", "Riya", "Karan"]
marks = [75, 90, 80]
plt.bar(names, marks)
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Student Performance")
plt.show()
```

Explanation:

- plt.bar() creates a bar chart.
- xlabel() and ylabel() define the axis labels.

- title() adds a heading to the graph.
- show() displays the visualization.

**4. Learning Outcomes**

- Understood the purpose and importance of Exploratory Data Analysis.
- Learned basic statistical measures for analysing datasets.
- Identified different types of graphs and their applications.
- Gained basic knowledge of the Matplotlib library.
- Practised creating graphical representations of data using Python.